

Math 526, S16

Quiz 11

Section:

Name: Answer

SSN: xxx-xx-

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer.

Problem 1. (10 points) Given a matrix $A = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}$.

(a) Find the eigenvalues λ 's and corresponding eigenvectors x 's of A .

$$\begin{aligned} \det(A - \lambda I) &= \det \begin{bmatrix} 1-\lambda & 2 \\ 5 & 4-\lambda \end{bmatrix} = (1-\lambda)(4-\lambda) - 10 \\ &= \lambda^2 - 5\lambda - 6 = 0 \\ &\Rightarrow (\lambda - 6)(\lambda + 1) = 0 \\ &\Rightarrow \lambda_1 = 6, \lambda_2 = -1 \end{aligned}$$

$$\begin{aligned} \text{For } \lambda_1 = 6, \text{ solve } (A - \lambda_1 I)x &= 0 \\ \begin{bmatrix} -5 & 2 \\ 5 & -2 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} &= 0 \Rightarrow x = c \begin{bmatrix} 2 \\ 5 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \text{For } \lambda_2 = -1, \text{ solve } (A - \lambda_2 I)x &= 0 \\ \begin{bmatrix} 2 & 2 \\ 5 & 5 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} &= 0 \Rightarrow x = c \begin{bmatrix} 1 \\ -1 \end{bmatrix} \end{aligned}$$

(b) Write an eigenvalue diagonalization $A = SAS^{-1}$ based on the results from (a).

$$\text{Let } S = \begin{bmatrix} 2 & 1 \\ 5 & -1 \end{bmatrix} \quad S^{-1} = \frac{1}{-7} \begin{bmatrix} -1 & -1 \\ -5 & 2 \end{bmatrix} = \begin{bmatrix} \frac{1}{7} & \frac{1}{7} \\ \frac{5}{7} & -\frac{2}{7} \end{bmatrix}$$

$$\Lambda = \begin{bmatrix} 6 & \\ & -1 \end{bmatrix} \Rightarrow A = S\Lambda S^{-1}$$